Computational Economics (2024) 63:1697–1704
https://doi.org/10.1007/s10614-024-10612-6

In Memoriam David A. Kendrick (1937–2024)

Co-founder of the Journal *Computational Economics*

Hans Amman[1] · Ruben Mercado[2] · Berç Rustem[3]

Published online: 25 May 2024

✉ Hans Amman
amman@uva.nl

Ruben Mercado
rubenmerc2@gmail.com

Berç Rustem
b.rustem@imperial.ac.uk

1 Faculty of Economics and Business, University of Amsterdam, Roetersstraat 11, 1018 WB Amsterdam, The Netherlands

2 United Nations Office, Buenos Aires, Argentina

3 Department of Computing, Imperial College, London, UK

1 In Loving Memory of David Kendrick

Our lifetime friend David Kendrick, born on November 14, 1937, in Gatesville, Texas, embarked on a remarkable journey that touched the hearts and minds of many in our community. With profound sadness, we bid farewell to a beloved husband, father, friend, and esteemed scholar who left an unforgettable mark on our lives. David passed away on April 7th, 2024 at the age of 86 in his hometown Austin.

1.1 The Academic

At the beginning of his career, David considered going into politics as he had been working as an assistant to Texas congressman W. R. Poage. Fortunately, for our community, David was more intrigued by science and felt less suited for a political career. After obtaining his Bachelor's degree at the University of Texas (UT) at Austin, David left Texas to pursue a Ph.D. at MIT (1966) that laid the groundwork for his later research interests in *Sectoral Economics and Planning* and *Optimal Control/optimization problems*, common in engineering. After finishing his Ph.D. he started working as an assistant professor at Harvard. (1966–1970).

Homeward-bound, David returned to his roots at the University of Texas as the Ralph W. Yarborough Centennial Professor of Liberal Arts in the early Seventies, where he began his groundbreaking research in stochastic control in (macro) economics. His passion for economic dynamics and control problems led him to serve as the founding editor of the *Journal of Economic Dynamics and Control*, He took great care in establishing the standing of the JEDC in the economics and engineering community. Under his editorial leadership JEDC attracted the attention of economists as well as engineers keen to contribute to the advancement of economic dynamics and the computational methods required to achieve this. In 2002 the JEDC published a special issue honoring David's work with contibutions of many well known economists.[1]

In 1981, David's seminal work, *Stochastic Control for Economic Models*,[2], illuminated the path forward for economists, laying the foundation for the importance of stochastic control theory in economics and the associated computational methods. This pivotal contribution showcased David's brilliance and set him apart as a visionary in his field. This was also evident in his contribution to the early development of the *GAMS* software, a pathbreaking software package that widened the scope and made work easier for generations of applied economists.

Throughout his illustrious career, David's dedication to advancing knowledge and fostering collaboration knew no bounds. As a co-founder of this journal *Computational Economics* in 1988, and later president of the *Society of Computational*

[1] David's contributions to the journal were celebrated in a Special issue of the *Journal of Economic Dynamics and Control* **26**, 2002. https://www.sciencedirect.com/science/article/abs/pii/S0165188901000744?via%3Dihub.

[2] Kendrick, D.A. *Stochastic Control for Economic Models* McGraw Hill Book Company, New York, 1981.

Economics, he played a central role in shaping the discourse of our discipline. The publication of the first *Handbook on Computational Economics*[3] in 1996 and undergraduate textbook *Computational Economics*,[4] in 2006, stand as a testament to his enduring legacy and his commitment to the field of computational economics.

To honor his contributions to the field, the Society of Computational Economics created the *David A. Kendrick Distinguished Service Award* to be given on an irregular basis to individuals who stand out for their contributions to the fields of computational economics and finance. The inaugural award was presented to David during a banquet at the Royal Society in London in July 2010.

Beyond his research achievements, David was known for his warmth, generosity, and outgoing kindness. As a teacher, he inspired generations of students to pursue their passions with fervor and dedication, leaving an indelible mark on their lives.

1.2 The Friend

To those who had the privilege of calling David a friend, his presence brought light and joy. For more than four decades, he graced our lives with his wisdom, humor, and continuing support.

David was a family man in the true sense of the word. Devoted to his wife Gail, proud of his children Ann and Colin, and as *Grandy*, he was the grandfather you hope for. Alongside his beloved wife, children, and grandchildren, David became an integral part of our families, enriching our lives with his boundless love and compassion. As an example, it spoke volumes about David's character that he took care of his chronically ill wife Gail unequivocally for many years while maintaining his cheerful demeanor. A demeanor that was supported by, our frequent visits to *Amy's icecream shops*, a local specialty in Texas. Many good ideas were conceived at Amy's.

It was very sad that David lost his daughter Ann about a year ago. A harsh life moment that feels so unfair. It was for us very encouraging to see that he spent the last years of his life in the good and loving care of his son Colin.

As we mourn the loss of our dear friend, let us remember David Kendrick not only for his scholarly achievements but also for his enduring kindness and generosity of spirit. To his son, grandchildren, and all who held him dear, we extend our deepest sympathies and heartfelt wishes during this difficult time.

Though David may not be with us, his legacy of love, knowledge, and compassion will continue to inspire us for generations to come. He will be profoundly missed, but his spirit will live on.

[3] *Handbook of Computational Economics* **1** , North-Holland, Amsterdam, 1996.

[4] *Computational Economics* Princeton University Press, 2006.

2 Selected Bibliography of David Kendrick's Work

2.1 Books

1. *Programming Investment in the Process Industries*, MIT Press, 1967.
2. *Notes and Problems in Microeconomic Theory*, (with S. Bowles). Markham Publishing Company, Chicago, Illinois, 1970.
3. *Frontiers in Quantitative Economics*, Volume II, (edited with M. Intriligator), North Holland Publishing Company, Amsterdam, 1974.
4. *The Planning of Industrial Programs: A Methodology* (with A. Stoutjesdijk), Johns Hopkins University Press, Baltimore, Maryland, 1978.
5. *Notes and Problems in Microeconomic Theory* (with P. Dixon and S. Bowles), North Holland Publishing Company, Amsterdam, 1980.
6. *La Programmation des Investissements Industriels* (with A. Choksi, A.Meeraus, and A.J. Stoutjesdijk), Economica, Paris, 1980.
7. *Stochastic Control for Economic Models*, McGraw Hill Book Company, New York, 1981. A free 2nd edition of the book can be obtained by contacting the corresponding author.
8. *The Planning of Investment Programs in the Steel Industry* (with A. Meeraus and J. Alatorre), The Johns Hopkins University Press, Baltimore, 1984.
9. *Feedback: A New Framework for Macroeconomic Policy*, Kluwer Publishing Company, Dordrecht, 1988. (Japanese language edition to published in 1989 by Koyo Shobo Publishers Ltd.)
10. *GAMS: Release 2.25: A Users Guide* (with A.Brooke and A. Meeraus), Scientific Press Series, Boyd and Fraser Publishing Co., Danvers, MA 01923, 1992.
11. *Models for Analyzing Comparative Advantage*, Kluwer Academic Publishers, Dordrecht, 1989.
12. *Handbook of Computational Economics* (edited with H.M Amman and J. Rust), North-Holland, Amsterdam, 1996.
13. *GAMS language guide* (with A. Brooke, A. Meeraus, R. Raman), GAMS Development Corporation, 2006.
14. *Computational Economics* (with P.R. Mercado and H.M. Amman), Princeton University Press, 2006.
15. *Feedback: a new framework for macroeconomic policy*, Springer Science & Business Media, 2012.

2.2 Articles and Contributions

16. Investment Planning and Economic Integration, *The Economics of Planning* **7**, 48–72, 1967.
17. A Dynamic Nonlinear Planning Model for Korea, (with L.Taylor), Practical Approaches to Development Planning by Irma Adelman (ed.), John Hopkins University Press, Baltimore, Maryland, 213–240, 1969.

18. Numerical Methods and Nonlinear Planning Models, (with L. Taylor), *Econometrica*, 453–467, 1970.
19. Optimal Operations of a System of Waste Water Treatment Facilities, (with H. S. Rao and C. H. Wells), *Proceedings of the 1970 IEEE Symposium on Adaptive Processes Decision and Control*, University of Texas at Austin, XII.3.1-XII.3.5, 1970.
20. Introduction to Part I, with Arthur MacEwan in *Studies in Development Planning* by Hollis Chenery (ed.), Harvard University Press, Cambridge, Mass, 7–10, 1971.
21. A Branch and Bound Algorithm for Zero–One Mixed Integer Programming Problems, (with R. Davis and M. Weitzman), *Operations Research* **19**, 1036–1044, 1971.
22. Mathematical Methods for Regional Planning, *Regional and Urban Economics* **1**, 1971.
23. Numerical Methods and Nonlinear Optimizing Models for Economic Planning, (with L. Taylor), in *Studies in Development Planning* by Hollis Chenery (ed.), Harvard University Press, Cambridge, Mass, 11–28, 1971.
24. Systems Problems in Economic Development, *Economic Analysis of Engineering and Social Systems* by J. Morley English (ed.), John Wiley and Sons, New York,183-234, 1972.
25. On the Leontief Dynamic Inverse, *Quarterly Journal of Economics* LXXXVI, 693–696, 1972.
26. Numerical Methods for Urban Planning, *Swedish Journal of Economics* **74**, 45–67, 1972.
27. Stochastic Control with Uncertain Macroeconomic Parameters (with J. Majors), *Automatica* **10**, 587–593, 1974.
28. Feedback Control Rule for Cocoa Market Stabilization (with H. Kim and L. Goreux) in *Quantitative Models of Commodity Markets*, by Walter C. Labys (ed.), Ballinger Books, Inc., Cambridge, Massachusetts, 1975, 233–264.
29. Control Theory and Economics (with M. Athans), *IEEE Transactions on Automatic Control* **AC-19**, 518-523, 1974.
30. Applications of Control Theory to Macroeconomics, *Annals of Economic and Social Measurement* **5**, 171–190, 1976.
31. Non-Convexities from Probing in Adaptive Control Problems, *Economic Letters* **1**, 347–351, 1978.
32. Adaptive Control of Macroeconomic Models with Measurement Error, in *Optimal Control of Econometric Models: An Approach to Economic Policy Formation*, by Holly, S., B. Rustem, and M. B. Zarrop (eds.), MacMillian Publishing Co., New York, New York, 204–227, 1979.
33. Control Theory: With Applications to Economics, Chapter 4 in K. J. Arrow and Michael D. Intriligator (eds.) *Handbook of Mathematical Economics*, North-Holland, Amsterdam, the Netherlands, 1981.
34. Some Properties of the Dual Adaptive Stochastic Control Algorithm, (with Pierre L. Dersin and Michael Athans), *IEEE Transactions on Automatic Control* **AC-26**, 1001-1008, 1981.

35. Caution and Probing in a Macroeconomic Model, *Journal of Economic Dynamics and Control* **4**, 149–170, 1982.
36. A Relational Database for the U.S. Economy, in Charles Kindleberger and Guido di Tella (eds.), Volume III of *Economics in the Long View: Essays in Honor of W. W. Rostow*, MacMillian, London, 1982.
37. Stochastic Control and Uncertainty in Dynamic Economic Systems, Chapter IX in M. Hazewinkel and A.H.G. Rinnooy Kan (eds), *Current Developments in the Interface: Economics, Econometrics, Mathematics*, Reidel Publishing Company, Dordrecht, the Netherlands, 1982.
38. Style in Multisectoral Modeling, Hughes Hallet, A. J. (ed), *Applied Decision Analysis and Economic Behavior*, Martinus Nijhoff Publishers, Dordrecht, 329–360, 1984.
39. A Mathematical-Computer Language for Linear Programming Problems, in T. Basar and L. F. Pau (eds), *Dynamic Modelling and Control of National Economies*, Pergamon Press Ltd., Oxford, United Kingdom, 233–240, 1984.
40. Controllo Stocastico con Apprendimento Passivo e Modelli Economici, *Rivista Internaxionale di Scienze Sociali* XCII, pp. 66–73, 1984.
41. The Tradeoff Between Economies of Scale and Reliability in the Electric Power Industry, (with Kwang Ha Kang), *Journal of Economic Development* **10**, 47-62, 1985.
42. Software for Economic Optimal Control Models, forthcoming in Sartore, D. and C. Carraro (eds.), *Developments of Control Theory for Economic Analysis*, Martinus Nijhoff Publishers, Dordrecht, 1986.
43. A Knowledge-Based System for Production and Distribution Economics, (with Ramayya Krishnan and Ronald M. Lee), *Computational Economics* **1**, 53-72, 1988.
44. A Production Model Construction System: Math Programming to Structured Modeling, in L. F. Pau, J. Motiwalla, Y. H. Pao and H. H. Teh (eds), *Expert Systems in Economics, Banking and Management*, North-Holland, Amsterdam, the Netherlands, 1989.
45. A Comparison of Structured Modeling and GAMS, (with R. Krishnan), *Computational Economics* **2**, 7, 1989.
46. A Production Model Construction System: PM Statement to Math Programming, *Journal of Economic Dynamics and Control* **14**, 219–236, 1990.
47. Parallel Model Representations, *Expert Systems with Applications* **1**, 383–389, 1990.
48. A Graphical Interface for Production and Transportation System Modeling: PTS, *Computational Economics* **4**, 229–236, 1991.
49. Parallel Processing for Large Scale Nonlinear Control Experiments in Economics, (with H.M. Amman), *International Journal of Supercomputer Applications* **5**, 90–95, 1992.
50. Uncertainty in Dynamic Macroeconomics: A Computational Challenge, in Billingsley, Keith R., Hilton U. Brown III and Ed Derohanes (eds), *Scientific Excellence in Supercomputing*, The IBM 1990 Contest Prize Papers, Ch. 21, 687–704, The Baldwin Press, The University of Georgia, Athens, Georgia, 1992.

51. Oil Refinery Modeling with the GAMS Language, (with A. Meeraus and Jung Sun Suh), Ch. 40 in Shri Bhagwan Dahiya (ed), *Theoretical Foundations of Development Planning*, Concept Publishing Co., New Delhi, India, 1992.
52. Active Learning: Monte Carlo Results (with H.M. Amman), *Journal of Economic Dynamics and Control* **18**, 119–124, 1994.
53. Nonconvexities in Stochastic Control Models: An Analysis (with H.M. Amman), in W. W. Cooper and A. B. Whinston (eds), *New Directions in Computational Economics*, Kluwer Academic Publishers, Dordrecht, The Netherlands, 57–94, 1994.
54. Solving Stochastic Optimization Models with Learning and Rational Expectations (with H.M. Amman and S. Achath), *Economics Letters* **48**, 9–13, 1995.
55. Ten Wishes, *Computational Economics* **8**, 67–80, 1995.
56. Nonconvexities in Stochastic Control Models (with H.M. Amman), *International Economic Review* **36**, 455-475, 1995.
57. Sectoral Economics, in *Handbook of Computational Economics* **1**, edited by Hans Amman, David Kendrick and John Rust, Chapter 6, 1996.
58. Forward Looking Variables in Deterministic Control (with H.M. Amman), *Annals of Operations Research* **68**, 141–159, 1996.
59. The DUALI/DUALPC Software for Optimal Control Models: Introduction (with H.M. Amman), in Hans Amman, Berc Rustem and Andrew Whinston, *Computational Approaches to Economic Problems*, Kluwer Academic Publishers, Dordrecht, Holland, 363–372, 1997.
60. Erratum to 'Active Learning: Monte Carlo Results', *Journal of Economic Dynamics and Control* **21**, 1613–1614, 1997.
61. Teaching Macroeconomics with GAMS (with P. R. Mercado and Hans M. Amman), *Computational Economics* **12**, 125–149, 1998.
62. Computing the Steady State of Linear Quadratic Optimization Models with Rational Expectations (with H.M. Amman), *Economics Letters* **58**, 185-191, 1998.
63. Programming Languages in Economics (with H.M. Amman), *Computational Economics* **14**, 151–181, 1999.
64. Computational Methods for Macro Policy Analysis: Hall and Taylor's Model in Duali (with P. R. Mercado), in A. Hughes Hallett and P. McAdam (eds), *Analysis in Macroeconomic Modelling*, Kluwer Academic Publishers, Boston and Dordrecht, 179–206., 1999.
65. Should Macroeconomic Policy Makers Consider Parameter Covariances (with H.M. Amman), *Computational Economics* **14**, 263–267., 1999
66. Linear Quadratic Optimization for Models with Rational Expectations (with Hans M. Amman), *Macroeconomic Dynamics* **3**, 534–543., 1999.
67. Matrix Methods for Solving Nonlinear Dynamic Optimization Models (with H.M. Amman), in R. J. Heijmans, D.S.G. Pollock and A. Satorra (eds), *Innovations in Multivariate Statistical Analysis*, A Festschrift for Heinz Neudecker, Kluwer Academic Publishers, Dordrecht, The Netherlands, 257–276, 2000.
68. Caution in Macroeconomic Policy: Uncertainty and the Relative Intensity of Policy (with P. R. Mercado), *Economics Letters* **68**, 37–41, 2000.

69. Stochastic policy design in a learning environment with rational expectations (with H.M. Amman), *Journal of Optimization Theory and Applications* **105**, 509–520, 2000.
70. Caution in macroeconomic policy: Uncertainty and the relative intensity of policy (with P.R. Mercado), *Economics Letters* **68**, 37–41,2000.
71. Mitigation of the Lucas critique with stochastic control methods (with H.M. Amman), *Journal of Economic Dynamics and Control* **27**, 2035–2057, 2003.
72. Stochastic control for economic models: past, present and the paths ahead, *Journal of economic dynamics and control* **29**, 3–30,2005.
73. Parameter Uncertainty and Policy Intensity: some extensions and suggestions for further work (with P.R. Mercado), *Computational Economics* **27**, 483–496,2006.
74. A classification system for economic stochastic control models (with H.M. Amman), *Computational Economics* **27**, 453–481,2006.
75. Computational economics: help for the underestimated undergraduate (with P.R. Mercado, H.M. Amman), *Computational Economics* **27**, 261–271, 2006.
76. Teaching computational economics to graduate students, *Computational Economics* **30**, 381–391,2007.
77. Solving the Beck and Wieland model with optimal experimentation in DUALPC (with H.M. Amman, M.P. Tucci), *Automatica* **44**, 1504–1510,2008.
78. The parameter set in an adaptive control Monte Carlo experiment: Some considerations (with M.P. Tucci and H.M. Amman), *Journal of Economic Dynamics and Control* **34**, 1531–1549,2010.
79. Expected optimal feedback with time-varying parameters (with M.P. and H.M. Amman), *Computational Economics* **42**, 351–371, 2013.
80. Quarterly fiscal policy, (with HM Amman), *The Economists' Voice* **11** (1), 7–12, 2014.
81. Quarterly fiscal policy experiments with a multiplier-accelerator model (with G. Shoukry), *Computational Economics* **43**, 269–293, 2014.
82. Comparison of policy functions from the optimal learning and adaptive control frameworks (with H.M. Amman), *Computational Management Science* **11**, 221–235, 2014.
83. Learning about learning in dynamic economic models (with HM Amman and MP Tucci), *Handbook of Computational Economics* **3**, 1–35, 2014.
84. Approximating the value function for optimal experimentation, (with H. M. Amman and M. P. Tucci) *Macroeconomic Dynamics* **24**, 1073–1086, 2020